

EBAU - 2022 EKAINA - BU

$$f(x) = e^x$$

$$g(x) = e^{-x}$$

$$h(x) = y = e$$

Bi funtzioen ebaketa puntua.

$$e^x = e^{-x}$$

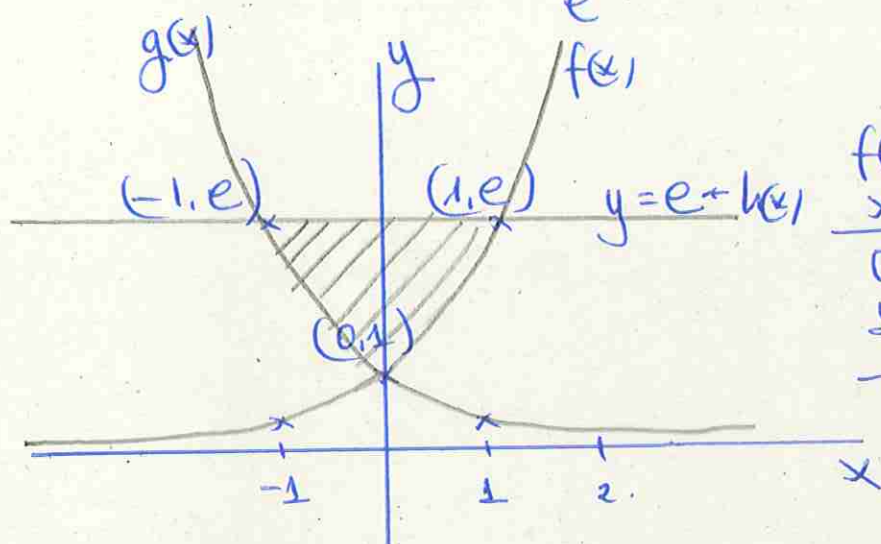
$$\frac{e^x}{e^{-x}} = 1$$

$$e^{x+x} = 1$$

$$2x = 0$$

$$x = 0 \cdot y = 1$$

$$P(0, 1)$$



$f(x)$	x	y
	0	1
	1	$e = 2,7$
	-1	$e^{-1} = 0,37$

$g(x)$	x	y
	0	1
	1	$e^{-1} = 0,37$
	-1	$e = 2,7$

$$A = \int_{-1}^0 (h(x) - g(x)) dx + \int_0^1 (h(x) - f(x)) dx$$

$$= 2 \int_{-1}^0 (h(x) - g(x)) dx = \boxed{2 \int_0^1 (h(x) - f(x)) dx}$$

$$A = 2 \int_0^1 (h(x) - f(x)) dx = 2 \int_0^1 (e - e^x) dx =$$

$$= 2 \left[ex - e^x \right]_0^1 = 2 \left[(e - e) - (0 - e^0) \right] =$$

$$= 2 \cdot 1 = \boxed{2}$$

$$f(x) = x$$

$$g(x) = x/8$$

$$h(x) = 1/x^2$$

Lehen kuadrantean.

Hiru funtzioen adierazpen grafikoa

→ $h(x) = 1/x^2$ Daxur = $\mathbb{R} - \{0\}$.
Simetrikoa da.

$$h(x) = \frac{1}{x^2}$$

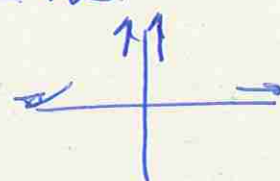
Asintotak - Bertikale $x=0$. $\lim_{x \rightarrow 0^+} \frac{1}{x^2} = +\infty$

Horizontale

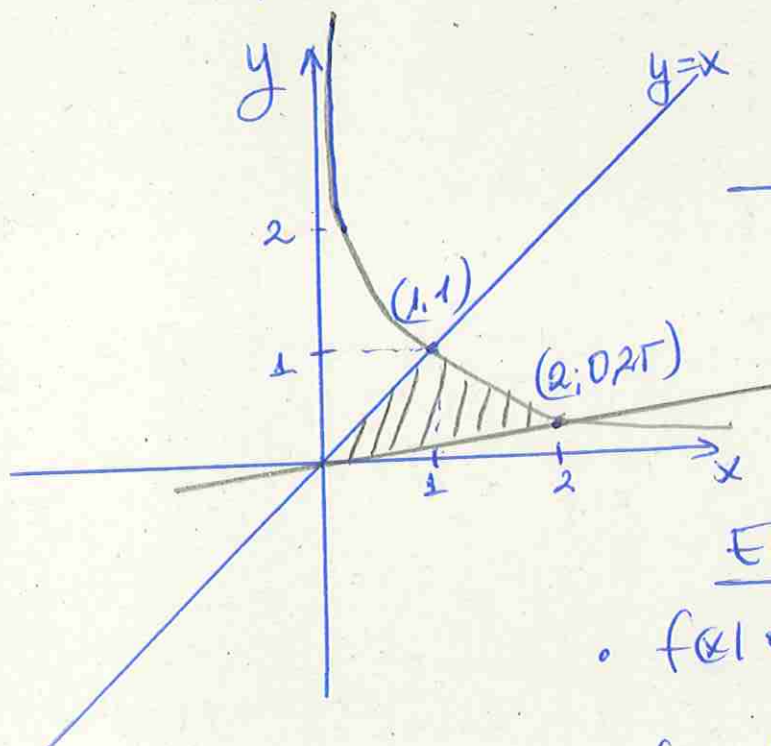
$$\lim_{x \rightarrow \infty} \frac{1}{x^2} = 0 \rightarrow y=0$$

$$x \rightarrow +\infty \quad \frac{1}{+} = 0^+ > 0 \text{ positiua}$$

$$x \rightarrow -\infty \quad \frac{1}{+} = 0^+ > 0 \text{ positiua}$$



x	y
1	1
2	0,25
-1	1
-2	0,25



$$\rightarrow f(x) = x$$

x	y
0	0
1	1
-1	-1

$$\rightarrow g(x) = \frac{x}{8}$$

x	y
0	0
8	1
1	8

Ebakets puntuok

• $f(x)$ eta $g(x)$, $x = \frac{x}{8}$ $\frac{x}{8} - x = 0$
 $-7/8 x = 0$

• $f(x)$ eta $h(x)$

$$\begin{cases} y = x \\ y = 1/x^2 \end{cases}$$

$$x = \frac{1}{x^2}$$

$$x^3 = 1 \quad \boxed{x=1, y=1}$$

• $g(x)$ eta $h(x)$

$$\begin{cases} y = x/8 \\ y = 1/x^2 \end{cases}$$

$$\frac{x}{8} = \frac{1}{x^2}$$

$$x=2 \quad y=0,25$$

$$A = \int_0^1 (f(x) - g(x)) dx + \int_1^2 (h(x) - g(x)) dx =$$

$$= \int_0^1 \left(x - \frac{x}{8}\right) dx + \int_1^2 \left(\frac{1}{x^2} - \frac{x}{8}\right) dx =$$

$$= \int_0^1 \frac{7x}{8} dx + \int_1^2 \left(x^{-2} - \frac{x}{8}\right) dx =$$

$$= \frac{7}{8} \frac{x^2}{2} \Big|_0^1 + \frac{x^{-1}}{-1} - \frac{x^2}{16} \Big|_1^2 = \frac{7}{16} + \left(-\frac{1}{2} - \frac{4}{16}\right) - \left(-1 - \frac{1}{16}\right)$$

$$= \frac{7}{16} + \frac{-8-4}{16} - \frac{-16-1}{16} = \frac{7}{16} + \frac{-12}{16} + \frac{17}{16} = \frac{7-12+17}{16} = \frac{12}{16}$$

$$= \boxed{\frac{3}{4} u^2}$$

$$f(x) = \frac{1}{x}$$

$$g(x) = x^2$$

$$h(x) = \frac{x^2}{8}$$

IRUDIKAPENA
 $f(x) = \frac{1}{x}$ Aldearabitek funtzioa

 $g(x) = x^2$ Parabola $\in (0, \infty)$
 $h(x) = x^2/8$ Parabola $\in (0, \infty)$

x	1	0.5	2
y	1	2	0.5

x	1	2
y	1	4

x	0	1	2
y	0	$\frac{1}{8}$	$\frac{1}{2}$

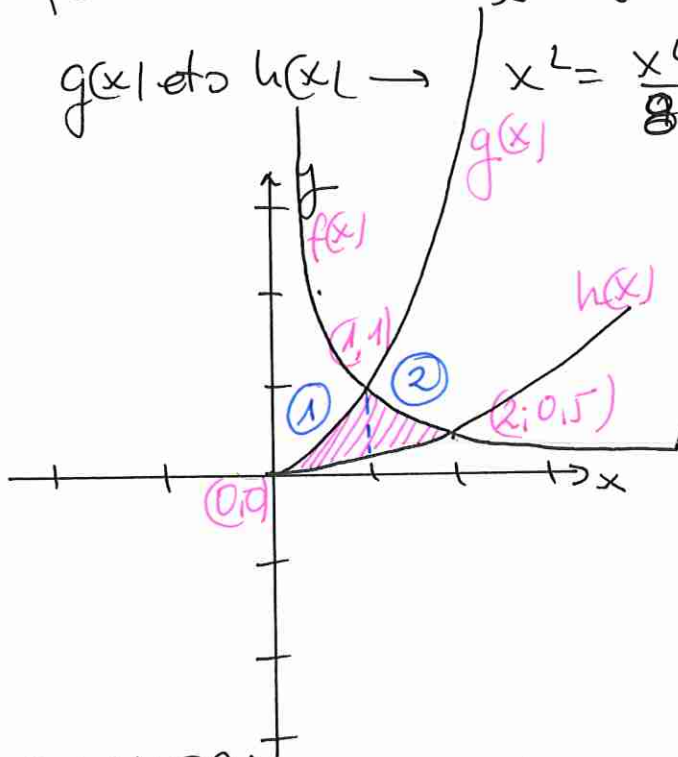
1.) EBAKI PUNTUAK.

$$f(x) \text{ eta } g(x) \rightarrow \frac{1}{x} = x^2 \quad x^3 = 1$$

$$x = 1 \quad y = 1 \quad \boxed{(1, 1)}$$

$$f(x) \text{ eta } h(x) \rightarrow \frac{1}{x} = \frac{x^2}{8} \rightarrow x^3 = 8 \quad \boxed{x = 2 \quad y = 0.5}$$

$$g(x) \text{ eta } h(x) \rightarrow x^2 = \frac{x^2}{8} \rightarrow \frac{7}{8}x^2 = 0 \quad \boxed{x = 0 \quad y = 0}$$

2.) Kalkula funtzioak

$$\textcircled{1} \text{ esparrua } g(x) - h(x) =$$

$$= x^2 - \frac{x^2}{8} = \frac{7}{8}x^2$$

$$\textcircled{2} \text{ esparrua } f(x) - h(x) =$$

$$= \frac{1}{x} - \frac{x^2}{8}$$

3. AZALERA

$$A = \int_0^1 \frac{7}{8}x^2 dx + \int_1^2 \left(\frac{1}{x} - \frac{x^2}{8} \right) dx =$$

$$= \left[\frac{7}{8} \frac{x^3}{3} \right]_0^1 + \left[\ln x - \frac{x^3}{24} \right]_1^2 = \frac{7}{24} + \left(\ln 2 - \frac{8}{24} \right) - \left(\ln 1 - \frac{1}{24} \right) =$$

$$= \frac{7}{24} + \ln 2 - \frac{8}{24} + \frac{1}{24} = \boxed{\ln 2 + \frac{1}{24}}$$

2021 @ OHIKOA AU

$$y = 4x - x^2$$

$$y = x^2 - 6$$

1) EBAKI PUNTUAK

$$4x - x^2 = x^2 - 6$$

$$0 = 2x^2 - 4x - 6$$

$$x = \frac{4 \pm \sqrt{4^2 - 4 \cdot 2 \cdot (-6)}}{2 \cdot 2} = \frac{4 \pm 8}{4}$$

$$x_1 = 3 \quad y_1 = 3$$

$$x_2 = -1 \quad y_2 = -5$$

$$\boxed{\begin{matrix} (3, 3) \\ (-1, -5) \end{matrix}}$$

Irudi kopuru

$$f(x) = 4x - x^2$$

$$\cap E(+2, 4)$$

Ox ardatzot:

$$0 = 4x - x^2$$

$$0 = x(4 - x)$$

$$x_1 = 0 \quad (0, 0)$$

$$x_2 = 4 \quad (4, 0)$$

$$g(x) = x^2 - 6$$

$$\cup E(0, -6)$$

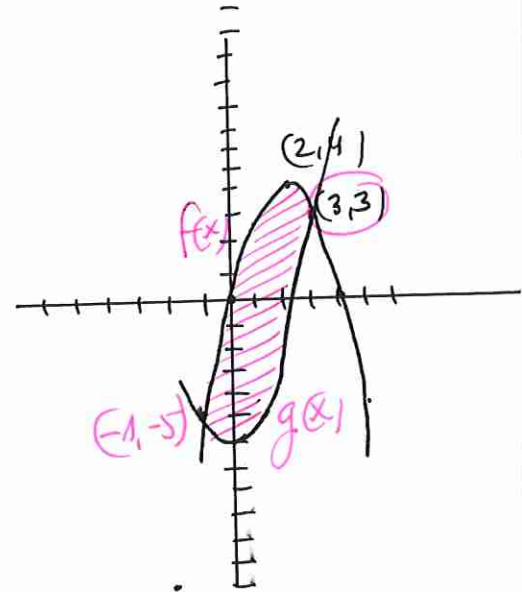
Ox ardatzot:

$$0 = x^2 - 6$$

$$x = \pm \sqrt{6}$$

$$(\sqrt{6}, 0)$$

$$(-\sqrt{6}, 0)$$



2.) Kurketo funtzioa

$$h(x) = f(x) - g(x) = (4x - x^2) - (x^2 - 6)$$

$$h(x) = -2x^2 + 4x + 6$$

3.) Azaleren
Barrowen erregelak $\int_{-1}^3 h(x) dx = H(3) - H(-1)$

$$A = \int_{-1}^3 (-2x^2 + 4x + 6) dx = \left[-\frac{2x^3}{3} + 2x^2 + 6x \right]_{-1}^3 =$$

$$= \left(-\frac{2 \cdot 3^3}{3} + 2 \cdot 3^2 + 6 \cdot 3 \right) - \left(-\frac{2 \cdot (-1)^3}{3} + 2 \cdot (-1)^2 + 6 \cdot (-1) \right) =$$

$$= (-18 + 18 + 18) - \left(\frac{2}{3} + 2 - 6 \right) = 18 - \frac{2}{3} + 4 =$$

$$= 22 - \frac{2}{3} = \boxed{\frac{64}{3} \text{ u}^2}$$

2020 OHIKOA A4

$$f(x) = x^2 - 2x + 1$$

$$g(x) = -x^2 + 5$$

1.) EBAKI-PUNTUAK

$$x^2 - 2x + 1 = -x^2 + 5$$

$$2x^2 - 2x - 4 = 0$$

$$x^2 - x - 2 = 0$$

$$x = \frac{1 \pm \sqrt{1 - 4(-2)}}{2}$$

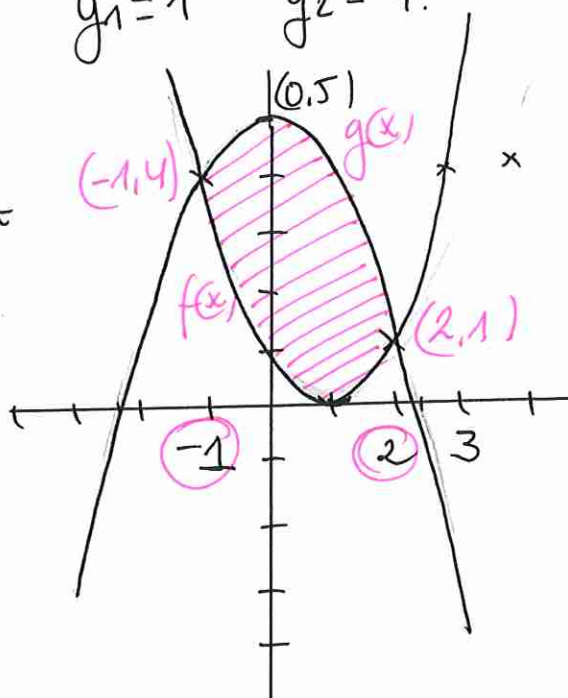
$$x = \frac{1 \pm 3}{2}$$

$$x_1 = 2$$

$$y_1 = 1$$

$$x_2 = -1$$

$$y_2 = 4$$



Irudikapena

$$f(x) = x^2 - 2x + 1$$

$$E(1, 0)$$

Ox ardatik (1, 0)

$$g(x) = -x^2 + 5$$

$$E(0, 5)$$

Ox ardatik (0, 5)

$$x_1 = \sqrt{5}$$

$$x_2 = -\sqrt{5}$$

2.) Kenketak funtzioa

barraio-berakoa

$$h(x) = g(x) - f(x)$$

$$h(x) = (-x^2 + 5) - (x^2 - 2x + 1)$$

$$h(x) = -2x^2 + 2x + 4$$

3.) Azalera

$$A = \int_{-1}^2 h(x) dx = H(2) - H(-1)$$

BARROWEN ERREJENUA

$$A = \int_{-1}^2 (-2x^2 + 2x + 4) dx = \left[-\frac{2x^3}{3} + x^2 + 4x \right]_{-1}^2 =$$

$$= \left(-\frac{2 \cdot 2^3}{3} + 2^2 + 4 \cdot 2 \right) - \left(-\frac{2(-1)^3}{3} + (-1)^2 + 4(-1) \right) =$$

$$= -\frac{16}{3} + 4 + 8 - \frac{2}{3} - 1 + 4 = 15 - \frac{18}{3} = 15 - 6 = \boxed{9}$$

2020 ET OIKOIA

A4

$$y = 3 - x^2$$

$$y = 2x$$

1. EBAK-PUNTNAK

$$3 - x^2 = 2x$$

$$0 = x^2 + 2x - 3$$

$$x = \frac{-2 \pm \sqrt{2^2 - 4 \cdot (-3)}}{2} = \frac{-2 \pm 4}{2}$$

$$x_1 = -3 \quad y_1 = -6$$

$$x_2 = 1 \quad y_2 = 2$$

$$\boxed{\begin{matrix} (-3, -6) \\ (1, 2) \end{matrix}}$$

Irudikopere

$$f(x) = 3 - x^2$$

$$E(0, 3)$$

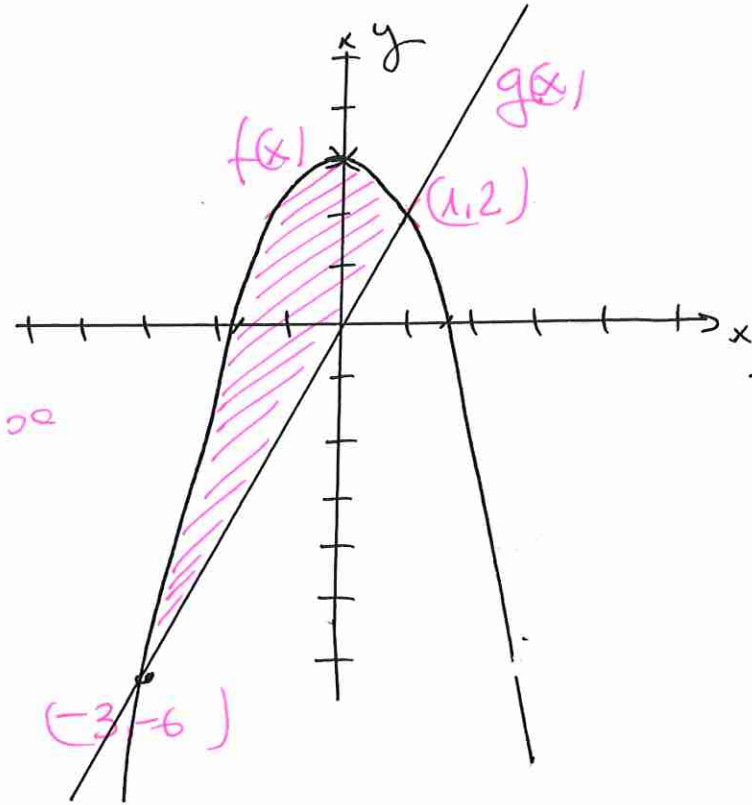
Ox ardothopet

$$0 = 3 - x^2$$

$$x = \pm \sqrt{3}$$

$$g(x) = 2x$$

x	y
0	0
1	2



2.) Kenketofuntzioa

Goko funtzioa - Beluk funtzioa

$$h(x) = f(x) - g(x) =$$

$$h(x) = (3 - x^2) - 2x \Rightarrow$$

$$h(x) = -x^2 - 2x + 3$$

3.) Azaketa

Barrowen erregelak

$$\int_{-3}^1 (-x^2 - 2x + 3) dx = H(1) - H(-3)$$

$$\int_{-3}^1 (-x^2 - 2x + 3) dx = \left[-\frac{x^3}{3} - x^2 + 3x \right]_{-3}^1 =$$

$$= \left(-\frac{1^3}{3} - 1^2 + 3 \cdot 1 \right) - \left(-\frac{(-3)^3}{3} - (-3)^2 + 3(-3) \right) =$$

$$= -\frac{1}{3} - 1 + 3 - \frac{27}{3} + 9 + 9 = -\frac{28}{3} + 20 =$$

$$= \boxed{\frac{32}{3}}$$

2019 EKAINA BY

$y = x + 2$ → zuzena irudikatzen da:

$y = x^2$

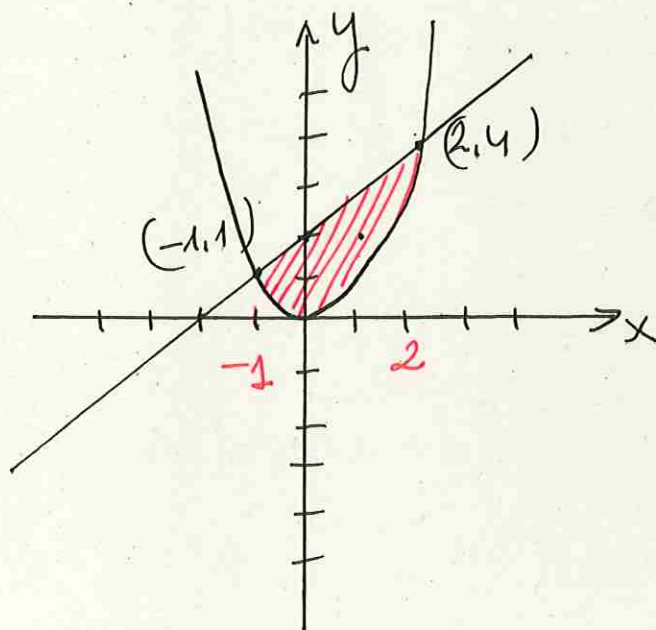
x	0	-2
y	2	0

→ PARABOLA ⇒

Erpinak (0,0)

Ox ardatza eta Oy ardatza (0,0).

x	0	1	2
y	0	1	4



1) Bi kurben arteko ebakidura puntuak:

$$x + 2 = x^2$$

$$x^2 - x - 2 = 0$$

$$x = \frac{1 \pm \sqrt{1 - 4(-2)}}{2} = \frac{1 \pm 3}{2}$$

$$x_1 = 2 \rightarrow y_1 = 4$$

$$x_2 = -1 \rightarrow y_2 = 1$$

goikoa - behekoa

2) keraketa funtzioa $h(x) = (x+2) - x^2$

3) azalera

$$A = \int_{-1}^2 h(x) dx = \int_{-1}^2 [(x+2) - x^2] dx =$$

Barrow

$$= \left[\frac{x^2}{2} + 2x - \frac{x^3}{3} \right]_{-1}^2 = \left(\frac{4}{2} + 2 \cdot 2 - \frac{2^3}{3} \right) - \left(\frac{(-1)^2}{2} + 2(-1) - \frac{(-1)^3}{3} \right)$$

$$= \left(2 + 4 - \frac{8}{3} \right) - \left(\frac{1}{2} - 2 + \frac{1}{3} \right) = 8 - \frac{1}{2} - \frac{9}{3} = \boxed{\frac{9}{2}}$$

2019 UZMAIA **[AU]**

$$y = x(3-x)$$
$$y = x^2$$

① $y = x(3-x)$
 $y = -x^2 + 3x$ PARABOLA PANBILA

Erpilo $x = -\frac{b}{2a} = \frac{-3}{2(-1)} = \frac{3}{2} = 1.5$

$f(3/2) = 2.25$ $E(1.5; 2.25)$

② $y = x^2$ ANURNA
 $E(0,0)$

0x ardatipst $(0,0)$ 0x ardatipst $0 = x(3-x)$ $\begin{cases} (0,0) \\ (3,0) \end{cases}$
0y ardatipst $(0,0)$

A) EBAKI PUNTNAK -bi kurben artaan

$$-x^2 + 3x = x^2$$

$$2x^2 - 3x = 0$$

$$x(2x-3) = 0 \quad \begin{cases} x_1 = 0 \\ x_2 = 3/2 \end{cases}$$

B) Kurketo funktsio

paikoo - Belukoo

$$x(3-x) - x^2$$

C) Azalevo

$$\int_{0.5}^{1.5} (x(3-x) - x^2) dx = \int_{0.5}^{1.5} (3x - x^2 - x^2) dx = \int_{0.5}^{1.5} (3x - 2x^2) dx =$$

$$= \left[\frac{3x^2}{2} - \frac{2x^3}{3} \right]_{0.5}^{1.5} = \left(\frac{3 \cdot (1.5)^2}{2} - \frac{2 \cdot (1.5)^3}{3} \right) - (0 - 0) = 1.125 u^2$$

$$= \underline{\underline{\frac{9}{8} u^2}}$$

